

WALMART

Black Friday Sales Analysis

Detailed Exploratory Data Analysis & Statistical Analysis Report

Project Type	Exploratory Data Analysis + Statistical Analysis
Transactions	550,068
Original Variables	10
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy
Statistical Method	95% Confidence Intervals / CLT-based analysis

Prepared from the supplied Walmart_Analysis.ipynb notebook.

1. Executive Summary

This project analyzes Walmart Black Friday transaction data to understand customer purchasing behaviour and identify patterns across demographic and customer attributes. The analysis examines gender, age, marital status, city category, occupation, product category, and purchase amount.

The dataset contains 550,068 transaction records across 10 original variables. The notebook performs data-quality checks, descriptive analysis, univariate and bivariate visualizations, outlier detection, correlation analysis, and confidence-interval analysis.

The analysis identifies clear descriptive differences in average purchase amounts between male and female customers, while the confidence intervals for married and unmarried customers overlap. Age-based analysis shows the 26–35 group as the largest customer segment, with its confidence interval overlapping the adjacent 36–50 group.

The project demonstrates how exploratory statistics and confidence intervals can be combined to move from simple descriptive observations toward more evidence-based customer analysis.

2. Business Problem & Objectives

The business objective is to understand how customer characteristics influence or relate to purchase amounts during Walmart's Black Friday sales period.

- Analyze the structure and characteristics of the Walmart transaction dataset.
- Understand customer distributions across gender, age, marital status, city category, and occupation.
- Examine the distribution of purchase amounts and identify unusual observations.
- Compare purchase behaviour across demographic groups.
- Use 95% confidence intervals to compare average spending between customer groups.
- Translate analytical findings into practical marketing and customer-segmentation recommendations.

3. Dataset Overview

Attribute	Value / Description
Transactions	550,068
Original columns	10
User_ID	Customer identifier
Product_ID	Product identifier
Gender	Male / Female
Age	7 age groups: 0–17 through 55+
Occupation	Codes 0–20
City_Category	A / B / C
Stay_In_Current_City_Years	Customer residence duration category
Marital_Status	0 / 1
Product_Category	Product category code
Purchase	Transaction purchase amount

The notebook creates an additional Age_Group field later in the analysis, bringing the working dataset to 11 columns.

4. Data Quality & Structure

- The dataset contains 550,068 transaction records.

- All 10 original columns are fully populated in the notebook's data-quality check.
- No missing values were found.
- Duplicate and unique-value checks were performed.
- Purchase is numeric and ranges from ■12 to ■23,961.
- Categorical attributes include Gender, Age, City_Category, Stay_In_Current_City_Years, and the later Age_Group field.

5. Descriptive Statistics

Purchase Metric	Value
Count	550,068
Mean	■9,263.97
Standard deviation	■5,023.07
Minimum	■12
25th percentile	■5,823
Median	■8,047
75th percentile	■12,054
Maximum	■23,961

- The mean purchase amount is approximately ■9,263.97.
- The median is ■8,047, below the mean, consistent with positive skew.
- The wide range indicates substantial variation in transaction values.

6. Customer Distribution Analysis

Gender

- Male customers: 414,259 transactions.
- Female customers: 135,809 transactions.
- Male customers form the larger transaction group.

Age

- The 26–35 age group has the highest transaction count.
- The dataset includes age groups from 0–17 through 55+.

City Category

- City B has the highest transaction representation, followed by City C and City A.

Marital Status

- Both married and unmarried customers are substantially represented.

Occupation

- Customers span occupation codes 0–20, indicating a diverse occupational mix.

7. Exploratory Data Analysis

The notebook uses histograms, countplots, boxplots, and a correlation heatmap to explore the data.

- **Purchase distribution:** positively skewed, with most purchases concentrated at lower-to-medium values and a smaller number of high-value transactions.
- **Gender distribution:** male transactions outnumber female transactions.
- **Age distribution:** 26–35 is the largest age segment.
- **Occupation:** customer volume varies considerably across occupation codes.
- **City:** City B has the highest representation.
- **Marital status:** married and unmarried customers are both well represented.
- **Purchase boxplot:** several high-value observations appear as outliers.

8. Purchase Behaviour by Customer Characteristics

Comparison	Observation from the analysis
Gender vs Purchase	Male customers have a higher average purchase than female customers.
Age vs Purchase	Customers in the 26–35 group generally show higher purchase amounts.
Marital Status vs Purchase	Spending patterns are similar between married and unmarried customers.
City Category vs Purchase	Purchase amounts vary slightly across city categories.

These are descriptive patterns from the supplied notebook. They should not be interpreted as causal effects without additional statistical modelling.

9. Gender Spending Analysis

The notebook directly addresses the question: **Are women spending more money per transaction than men?**

Gender	Average Purchase	95% Confidence Interval
Male	■9,437.53	■9,422.02 – ■9,453.03
Female	■8,734.57	■8,709.21 – ■8,759.92

The observed difference in average transaction value is approximately **■702.96**, with male customers showing the higher mean in this dataset.

The calculated 95% confidence intervals do not overlap. The notebook interprets this as evidence of a clear difference in average spending between the two groups. For rigorous statistical inference, a formal hypothesis test would be an appropriate additional step.

Business implication: differentiated customer engagement strategies can be explored, but should be validated against campaign response, profitability, and additional customer attributes.

10. Marital Status Confidence-Interval Analysis

Group	95% Confidence Interval
Married	■9,240.46 – ■9,281.89
Unmarried	■9,248.62 – ■9,283.20

The confidence intervals overlap. The notebook therefore concludes that married and unmarried customers have similar average spending behaviour.

- The analysis does not support creating separate pricing or promotional strategies based solely on marital status.
- Other dimensions such as age, product preference, shopping history, and customer segmentation may be more useful for targeted marketing.

11. Age-Based Confidence-Interval Analysis

For the statistical comparison, the notebook groups the original age categories into five groups: 0–17, 18–25, 26–35, 36–50, and 51+. The analysis calculates a 95% confidence interval for each group's mean purchase.

Adjacent age groups	Result
0–17 vs 18–25	Do not overlap
18–25 vs 26–35	Do not overlap
26–35 vs 36–50	Overlap
36–50 vs 51+	Do not overlap

The notebook notes that the 26–35 group's interval overlaps with the adjacent 36–50 group. Because the notebook checks adjacent groups rather than every possible pair, this conclusion should be understood in that scope.

12. Outlier Analysis

The notebook applies the Interquartile Range (IQR) method to the Purchase variable.

Measure	Result
Q1	■5,823
Q3	■12,054
IQR	■6,231
Outliers detected	2,677
Outlier percentage	0.49%

The analysis retains these observations because they are treated as genuine high-value Black Friday purchases rather than data-entry errors. For a sales-analysis use case, these transactions may contain useful business information.

13. Correlation Analysis

A correlation heatmap is used to inspect linear relationships among numerical variables. The notebook reports no strong linear correlation among the variables and identifies Product Category as having the strongest association with Purchase.

Correlation should be interpreted as association rather than causation. A weak or moderate correlation does not imply that a variable has no business value; relationships may be nonlinear or influenced by other variables.

14. Business Insights

- The dataset is complete and suitable for exploratory analysis because no missing values were identified.
- Purchase amounts are positively skewed, with a smaller number of high-value transactions.
- Male customers form the larger transaction group and have a higher average purchase than female customers in this dataset.
- The 26–35 age group is the largest customer segment.
- City Category B has the highest customer representation.
- Married and unmarried customers have overlapping confidence intervals, indicating similar average spending in this analysis.
- Age-group confidence intervals show differences between several adjacent groups, while 26–35 and 36–50 overlap.
- Only 0.49% of transactions are identified as Purchase outliers using the IQR rule.

15. Recommendations

- Continue analyzing the 26–35 segment because it is the largest customer group and represents an important audience for promotions.
- Use customer age and shopping preferences to create more relevant promotional campaigns rather than relying on one-size-fits-all marketing.
- Consider gender-based promotions where supported by campaign response data, since male customers have a higher average transaction value in this dataset.
- Avoid using marital status alone as a basis for pricing or promotion decisions because the confidence intervals overlap.
- Use purchase history and product preferences to improve product recommendations and customer engagement.
- Tailor inventory and promotions across city categories based on local demand patterns.
- Monitor high-value transactions to identify potential premium customer segments.
- Repeat the analysis around major shopping events to improve marketing and inventory planning.

16. Methodology

Stage	Approach
Data loading	Imported Walmart transaction data into Pandas.
Data inspection	Checked shape, information, descriptive statistics, duplicates, and unique values.
EDA	Analyzed categorical distributions and purchase behaviour using plots.
Bivariate analysis	Compared Purchase against Gender, Age, Marital Status, and City Category.
Correlation	Generated a correlation heatmap for numerical variables.
Outliers	Applied the IQR rule to Purchase.
Statistical analysis	Calculated 95% confidence intervals for gender, marital status, and age groups.
Business interpretation	Converted observed patterns into customer and marketing recommendations.

17. Limitations & Future Scope

Limitations

- The dataset is transaction-level and does not by itself establish long-term customer behaviour.
- The analysis is primarily descriptive; formal hypothesis tests are not included in the supplied notebook.
- Confidence-interval overlap is used as the comparison criterion and should not be treated as a substitute for a formal significance test.
- Occupation and product-category values are coded, limiting direct business interpretation without a corresponding codebook.
- The analysis does not include profit, discount, marketing cost, or customer-level lifetime value measures.

Future Scope

- Perform formal two-sample hypothesis testing for gender and marital-status comparisons.
- Use ANOVA or suitable non-parametric tests for multi-group age comparisons.
- Analyze product-category level purchase and revenue patterns in greater depth.
- Create customer-level aggregates for segmentation and repeat-purchase analysis.
- Build RFM or customer-value segments.

- Develop a predictive model for purchase amount or customer/product behaviour.
- Create an interactive Power BI dashboard for stakeholder use.

18. Conclusion

The Walmart Black Friday analysis demonstrates an end-to-end exploratory and statistical workflow using Python. The project begins with data-quality assessment, progresses through descriptive and visual analysis, and then applies confidence intervals to compare spending patterns across customer groups.

The strongest practical findings are the higher average purchase observed for male customers, the prominence of the 26–35 age group, the high representation of City B, and the similarity in spending between married and unmarried customers based on overlapping confidence intervals.

The analysis provides a strong foundation for customer segmentation and targeted marketing. The next step would be to strengthen the statistical evidence with formal hypothesis testing and expand the analysis with product-level, customer-level, profitability, and predictive perspectives.

Appendix — Key Values

Metric	Value
Rows	550,068
Original columns	10
Unique users	5,891
Unique products	3,631
Gender categories	2
Age categories	7
Occupation categories	21
City categories	3
Product categories	20
Average purchase	■9,263.97
Median purchase	■8,047
Purchase standard deviation	■5,023.07
Purchase minimum	■12
Purchase maximum	■23,961
Purchase outliers	2,677 (0.49%)
Male average purchase	■9,437.53
Female average purchase	■8,734.57
Married 95% CI	■9,240.46 – ■9,281.89
Unmarried 95% CI	■9,248.62 – ■9,283.20

Source: Supplied Walmart_Analysis.ipynb. This report follows the analysis and terminology in the notebook and does not add unsupported dataset findings.

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